

B28: SQL SuperHero Program : Session 19

1 message

Vishal Jaiswal <vishaldbdev@gmail.com>
Bcc: satishhattekar1997@gmail.com

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Questions from Session, I discussed

find for each stock, buying & selling at what price would have generated maximum profit.

```
CREATE TABLE stocks(
stock_id int,
stock_name varchar(50),
stock_date date,
stock_price decimal(10,2)
);
```

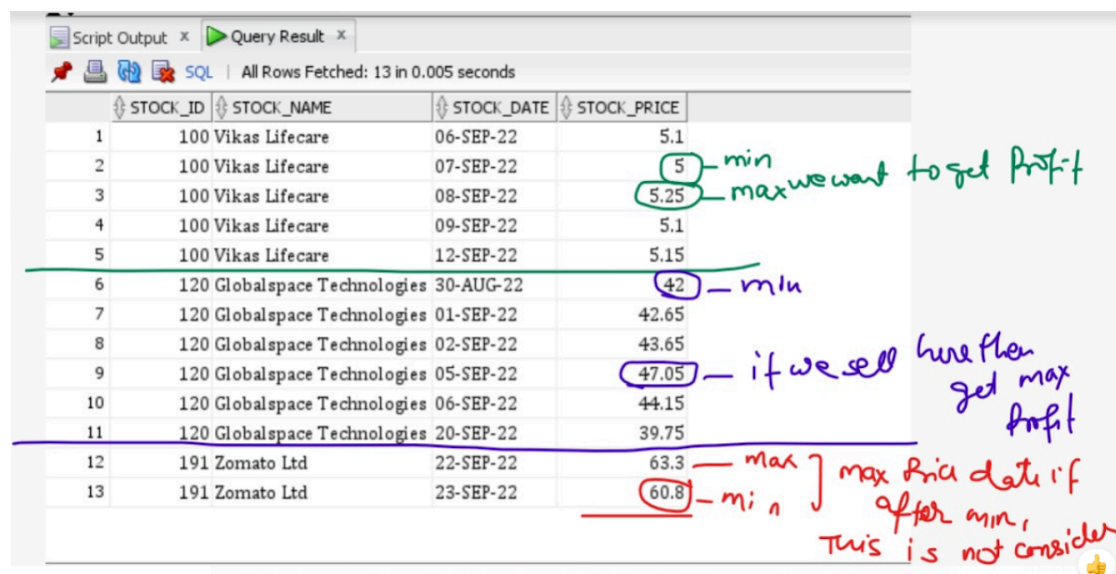
```
INSERT INTO stocks VALUES(100,'Vikas Lifecare','06-SEP-2022',5.10);
INSERT INTO stocks VALUES(100,'Vikas Lifecare','07-SEP-2022',5.00);
INSERT INTO stocks VALUES(100,'Vikas Lifecare','08-SEP-2022',5.25);
INSERT INTO stocks VALUES(100,'Vikas Lifecare','09-SEP-2022',5.10);
INSERT INTO stocks VALUES(100,'Vikas Lifecare','12-SEP-2022',5.15);
INSERT INTO stocks VALUES(120,'Globalspace Technologies','30-AUG-2022',42.00);
INSERT INTO stocks VALUES(120,'Globalspace Technologies','01-SEP-2022',42.65);
INSERT INTO stocks VALUES(120,'Globalspace Technologies','02-SEP-2022',43.65);
INSERT INTO stocks VALUES(120,'Globalspace Technologies','05-SEP-2022',47.05);
INSERT INTO stocks VALUES(120,'Globalspace Technologies','06-SEP-2022',44.15);
INSERT INTO stocks VALUES(120,'Globalspace Technologies','20-SEP-2022',39.75);
INSERT INTO stocks VALUES(191,'Zomato Ltd','22-SEP-2022',63.30);
INSERT INTO stocks VALUES(191,'Zomato Ltd','23-SEP-2022',60.80);
```

Output contains 2 rows

1. Vikas lifecare / bought on 07-SEP-2022 in \$5 an sold on 08-SEP-2022 in 5.25/ profile 0.25
2. Globalspace technology bought 30-AUG-2022 in 42 and sold 05-SEP-2022 in 47.05

=====

zomato will not come as it minimum price on 23-sep and after that after that price not increased. Globalspace technology have minimum price as 39 on 20-SEP but after that price not incresed, so will not consider that. we will consider 42 price on 30-AUG becaus ethen we able to sell on 05-sep for maximum profit



STOCK_ID	STOCK_NAME	STOCK_DATE	STOCK_PRICE
1	100 Vikas Lifecare	06-SEP-22	5.1
2	100 Vikas Lifecare	07-SEP-22	5
3	100 Vikas Lifecare	08-SEP-22	5.25
4	100 Vikas Lifecare	09-SEP-22	5.1
5	100 Vikas Lifecare	12-SEP-22	5.15
6	120 Globalspace Technologies	30-AUG-22	42
7	120 Globalspace Technologies	01-SEP-22	42.65
8	120 Globalspace Technologies	02-SEP-22	43.65
9	120 Globalspace Technologies	05-SEP-22	47.05
10	120 Globalspace Technologies	06-SEP-22	44.15
11	120 Globalspace Technologies	20-SEP-22	39.75
12	191 Zomato Ltd	22-SEP-22	63.3
13	191 Zomato Ltd	23-SEP-22	60.8

Handwritten annotations:

- For Vikas Lifecare: 5 (min), 5.25 (max we want to get Profit)
- For Globalspace Technologies: 42 (min), 47.05 (if we sell here then get max Profit)
- For Zomato Ltd: 60.8 (min), 63.3 (max Price date if after min, This is not consider)

basically what we want lets first translate. first do translation, requirement to english and then englisg to sql. in our english language we have to find out the stocks we are first purchased on a price and the sold. means sell_price must be more tahn buy price and on that date definately sell date is

also more tahn buy date.

```
select buy.stock_id,
       buy.stock_name,
       buy.stock_date buy_date,
       buy.stock_price buy_price,
       sell.stock_date sell_date,
       sell.stock_price sell_price from stocks buy, stocks sell
       where buy.stock_id = sell.stock_id
       AND sell.stock_date > buy.stock_date
       AND sell.stock_price > buy.stock_price;
```

simple.. where sell_date is more than buy_date and sell price also more than buy price

Query Result:

STOCK_ID	STOCK_NAME	BUY_DATE	BUY_PRICE	SELL_DATE	SELL_PRICE
1	100 Vikas Lifecare	06-SEP-22	5.1	08-SEP-22	5.25
2	100 Vikas Lifecare	07-SEP-22	5.1	08-SEP-22	5.25
3	100 Vikas Lifecare	07-SEP-22	5.1	09-SEP-22	5.1
4	100 Vikas Lifecare	06-SEP-22	5.1	12-SEP-22	5.15
5	100 Vikas Lifecare	07-SEP-22	5.1	12-SEP-22	5.15
6	100 Vikas Lifecare	09-SEP-22	5.1	12-SEP-22	5.15
7	120 Globalspace Technologies	30-AUG-22	42.65	01-SEP-22	42.65
8	120 Globalspace Technologies	30-AUG-22	42.65	02-SEP-22	43.65
9	120 Globalspace Technologies	01-SEP-22	42.65	02-SEP-22	43.65
10	120 Globalspace Technologies	30-AUG-22	42.65	05-SEP-22	47.05
11	120 Globalspace Technologies	01-SEP-22	42.65	05-SEP-22	47.05
12	120 Globalspace Technologies	02-SEP-22	43.65	05-SEP-22	47.05
13	120 Globalspace Technologies	30-AUG-22	42.65	06-SEP-22	44.15
14	120 Globalspace Technologies	01-SEP-22	42.65	06-SEP-22	44.15
15	120 Globalspace Technologies	02-SEP-22	43.65	06-SEP-22	44.15

All data here now

- just play with english and then convert in sql. Now we have to pick the difference between sell and buy and where is the highest. because we have to make maximum profit. simple. so for EACH stock, we have to find out max difference
- each means --> group by. (discussed 1000 times the logic, hope u all know) but i was all column in output. (preserve input), so group by chnage to partition by
- Just add this below :
- **ROW_NUMBER() OVER(PARTITION BY buy.stock_id ORDER BY sell.stock_price - buy.stock_price DESC)**
- PARTITION BY buy.stock_id --> for each stock
- ORDER BY sell.stock_price - buy.stock_price DESC --> arrange as per difference between sell - buy in desc order
- ROW_NUMBER --> analytical function.. assign numbers

Query Result:

STOCK_ID	STOCK_NAME	BUY_DATE	BUY_PRICE	SELL_DATE	SELL_PRICE	RN
1	100 Vikas Lifecare	07-SEP-22	5.1	08-SEP-22	5.25	1
2	100 Vikas Lifecare	06-SEP-22	5.1	08-SEP-22	5.25	2
3	100 Vikas Lifecare	07-SEP-22	5.1	12-SEP-22	5.15	3
4	100 Vikas Lifecare	07-SEP-22	5.1	09-SEP-22	5.1	4
5	100 Vikas Lifecare	06-SEP-22	5.1	12-SEP-22	5.15	5
6	100 Vikas Lifecare	09-SEP-22	5.1	12-SEP-22	5.15	6
7	120 Globalspace Technologies	30-AUG-22	42.65	05-SEP-22	47.05	1
8	120 Globalspace Technologies	01-SEP-22	42.65	05-SEP-22	47.05	2
9	120 Globalspace Technologies	02-SEP-22	43.65	05-SEP-22	47.05	3
10	120 Globalspace Technologies	30-AUG-22	42.65	06-SEP-22	44.15	4
11	120 Globalspace Technologies	30-AUG-22	42.65	02-SEP-22	43.65	5
12	120 Globalspace Technologies	01-SEP-22	42.65	06-SEP-22	44.15	6
13	120 Globalspace Technologies	01-SEP-22	42.65	02-SEP-22	43.65	7

stock chage start again

DON (Amirah bachan) -> Ex cute first (what boundary)

then arrange per date

stock chage

start again

now just choose, where rn = 1. because it has the max profitable difference.

```

with n1 as (
Select buy.stock_id, buy.stock_name, buy.stock_date buy_date, buy.stock_price buy_price, sell.stock_date sell_date, sell.stock_price sell_price ,
ROW_NUMBER()
OVER(PARTITION BY buy.stock_id ORDER BY sell.stock_price - buy.stock_price DESC) rn
from stocks buy, stocks sell
where buy.stock_id = sell.stock_id
AND sell.stock_date > buy.stock_date
AND sell.stock_price > buy.stock_price
) select * from n1 where rn = 1

```

Added this

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.007 seconds

STOCK_ID	STOCK_NAME	BUY_DATE	BUY_PRICE	SELL_DATE	SELL_PRICE	RN
1	100 Vikas Lifecare	07-SEP-22	5	08-SEP-22	5.25	1
2	120 Globalspace Technologies	30-AUG-22	42	05-SEP-22	47.05	1

output

Another approach, as someone discussed in class that Can we do Via Group By

Whatever method you want, you can do and solve problem. This is the beauty of **WITH clause**, write your own code and combine together

```

With n1 as (
select buy.STOCK_ID, buy.STOCK_NAME, buy.STOCK_DATE buy_date, buy.STOCK_PRICE buy_price, sell.STOCK_DATE sell_date, sell.STOCK_PRICE sell_price from
stocks buy, stocks sell
where buy.STOCK_ID = sell.stock_id
and sell.STOCK_DATE > buy.STOCK_DATE
and sell.STOCK_PRICE > buy.STOCK_PRICE
) select * from n1

```

→ My 1st query gives me data

STOCK_ID	STOCK_NAME	BUY_DATE	BUY_PRICE	SELL_DATE	SELL_PRICE
100	Vikas Lifecare	06-SEP-22	5.1	08-SEP-22	5.25
100	Vikas Lifecare	07-SEP-22	5	08-SEP-22	5.25
100	Vikas Lifecare	07-SEP-22	5	09-SEP-22	5.1
100	Vikas Lifecare	06-SEP-22	5.1	12-SEP-22	5.15
100	Vikas Lifecare	07-SEP-22	5	12-SEP-22	5.15
100	Vikas Lifecare	09-SEP-22	5.1	12-SEP-22	5.15
120	Globalspace Technologies	30-AUG-22	42	01-SEP-22	42.65
120	Globalspace Technologies	30-AUG-22	42	02-SEP-22	43.65
120	Globalspace Technologies	01-SEP-22	42.65	02-SEP-22	43.65
120	Globalspace Technologies	30-AUG-22	42	05-SEP-22	47.05

On top of the above out put , just write group by as below:

```

With n1 as (
select buy.STOCK_ID, buy.STOCK_NAME, buy.STOCK_DATE buy_date, buy.STOCK_PRICE buy_price, sell.STOCK_DATE sell_date, sell.STOCK_PRICE sell_price, (sell.STOCK_PRICE-buy.STOCK_PRICE) diff from
stocks buy, stocks sell
where buy.STOCK_ID = sell.stock_id
and sell.STOCK_DATE > buy.STOCK_DATE
and sell.STOCK_PRICE > buy.STOCK_PRICE
)
select stock_name, max(DIFF) max_diff from n1 group by stock_name

```

→ n1 (My first query)

STOCK_NAME	MAX_DIFF
Vikas Lifecare	.25
Globalspace Technologies	5.05

2 rows selected.

got the max diff for each stock

Now you have N1 and You have N2. just combine both and get all data.

```
With n1 as (  
  select buy.STOCK_ID, buy.STOCK_NAME, buy.STOCK_DATE buy_date, buy.STOCK_PRICE buy_price, sell.STOCK_DATE sell_date, sell.STOCK_PRICE sell_price, (sell.STOCK_PRICE-buy.STOCK_PRICE) diff from  
    stocks buy, stocks sell  
  where buy.STOCK_ID = sell.stock_id  
    and sell.STOCK_DATE > buy.STOCK_DATE  
    and sell.STOCK_PRICE > buy.STOCK_PRICE  
), n2 as (  
  select stock_name, max(DIFF) max_diff from n1 group by stock_name  
)  
select n1.*  
from n1, n2  
where n1.STOCK_NAME = n2.STOCK_NAME  
and n1.DIFF = n2.max_diff
```

Handwritten notes: "1st" points to the first CTE, "2nd" points to the second CTE, and "combine" points to the final join query.

STOCK_ID	STOCK_NAME	BUY_DATE	BUY_PRICE	SELL_DATE	SELL_PRICE	DIFF
100	Vikas Lifecare	07-SEP-22	5	08-SEP-22	5.25	.25
120	Globalspace Technologies	30-AUG-22	42	05-SEP-22	47.05	5.05

2 rows selected.

```
CREATE TABLE employee (  
  empid INT PRIMARY KEY,  
  name VARCHAR(50),  
  salary DECIMAL(10, 2)  
);
```

```
INSERT INTO employee (empid, name, salary) VALUES  
(1, 'Alice', 60000.00),  
(2, 'Bob', 55000.00),  
(3, 'Charlie', 70000.00),  
(4, 'Diana', 65000.00),  
(5, 'Eve', 62000.00);
```

Problem: Write an SQL query to get empid, name, and salary of employees having the minimum salary difference.

```
CREATE TABLE employee (empid Number,name VARCHAR2(50),salary number);
```

```
INSERT INTO employee (empid, name, salary) VALUES (1, 'Alice', 60000);  
INSERT INTO employee (empid, name, salary) VALUES (2, 'Bob', 55000);  
INSERT INTO employee (empid, name, salary) VALUES (3, 'Charlie', 70000);  
INSERT INTO employee (empid, name, salary) VALUES (4, 'Diana', 65000);  
INSERT INTO employee (empid, name, salary) VALUES (5, 'Eve', 62000);
```

Problem Statement: What the question wants? we want to find out the employees who have the minimum salary difference. So, how we get it in plain english. We have to compare each employee salary with every other employee.

There are 2 words . 1. compare (whenever compare comes means lead and lag)

2. compaby each employee with every other, means all permutations (whenever all permutations means cross join)

--> This is the trick. Always try to convert requirement to english and english to sql

1. Use Cross Join

```
select e1.empid as e1_empid, e1.name as e1_name, e1.salary as e1_salary,  
       e2.name as e2_name, e2.salary as e2_salary, abs(e1.salary - e2.salary) as diff  
from employee e1, employee e2
```

This is cross join. we did same question previously also. (Maximum number of distinct matches). The above query will give you output with all emp e1 records match with all emp e2 records. But we can not compare same employees, so add one extra check as `e1.empid <> e2.empid`

```

1 select e1.empid as e1_empid, e1.name as e1_name, e1.salary as e1_salary,
2       e2.name as e2_name, e2.salary as e2_salary, abs(e1.salary - e2.salary) as diff
3 from employee e1, employee e2
4 where e1.empid <> e2.empid
5 order by diff

```

used it to consider + or - both same
arrange

we want this view

E1_EMPID	E1_NAME	E1_SALARY	E2_NAME	E2_SALARY	DIFF
5	Eve	62000	Alice	60000	2000
1	Alice	60000	Eve	62000	2000
5	Eve	62000	Diana	65000	3000
4	Diana	65000	Eve	62000	3000
1	Alice	60000	Diana	65000	5000

$3-2 = 1$
 $2-3 = -1$
abs will give 1
beoz diff is 1

--> I haven't taught fetch and rownum yet, else you can directly use fetch first 1 rows only and you get the data. But till now, I taught you `row_number()`. assign that and choose where `rn = 1`

2. Use Analytical.

Lets focus on compare keyword. compare means Analytical. Lead and Lag. Lets use and try to arrange the salary and compare them.

```

1 select empid, name, salary, lead(salary,1) over(order by salary) next_highest_sal
2 from employee
3
4

```

Compare next
comparing sal
Arrange first

EMPID	NAME	SALARY	NEXT_HIGHEST_SAL
2	Bob	55000	60000
1	Alice	60000	62000
5	Eve	62000	65000
4	Diana	65000	70000
3	Charlie	70000	-

Now just get the difference . between both and the least one is the answer

```

1 with N1 as (
2 select empid, name, salary, lead(salary,1) over(order by salary) next_highest_sal
3 from employee
4 ) select N1.*, (next_highest_sal-salary) as diff from n1 order by diff
5

```

Choose this one ✓

EMPID	NAME	SALARY	NEXT_HIGHEST_SAL	DIFF
1	Alice	60000	62000	2000
5	Eve	62000	65000	3000
4	Diana	65000	70000	5000
2	Bob	55000	60000	5000
3	Charlie	70000	-	-

=====

```
Create table emp (Empid Number, plancd varchar2(20), planstartdate date, planenddate date);
```

```
Insert Into Emp Values (101, 'A', '01-JAN-2024','31-JUL-2024');
Insert Into Emp Values (101, 'B', '01-APR-2024','31-DEC-2024');
Insert Into Emp Values (102, 'A', '01-JAN-2024','31-DEC-2024');
Insert Into Emp Values (102, 'B', '01-JAN-2025','31-JUL-2025');
Insert Into Emp Values (103, 'A', '01-JAN-2024','31-JUL-2024');
Insert Into Emp Values (103, 'B', '01-AUG-2024','31-MAR-2025');
Insert Into Emp Values (103, 'C', '01-JAN-2025','31-JUL-2025');
```

Write a SQL query to find out all such employees who have overlap.

Solution: If you see the question and the data, we can see that employee 101 have 2 rows the first rows Planenddate = 31-JUL-2024 and the 2nd row planstartdate = 01-APR-2024. It means from april to july, he worked on 2 plancode. means he has overlap.

Employee 102 has 2 rows, but we can not see any overlap.

Employee 103 has 3 rows and last 2 rows has overlap.

He has overlap

EMPID	PLANCD	PLANSTARTDATE	PLANENDDATE
101	A	01-JAN-24	31-JUL-24
101	B	01-APR-24	31-DEC-24
102	A	01-JAN-24	31-DEC-24
102	B	01-JAN-25	31-JUL-25
103	A	01-JAN-24	31-JUL-24
103	B	01-AUG-24	31-MAR-25
103	C	01-JAN-25	31-JUL-25

7 rows selected.

no overlap

overlap

How can we get overlap? If we see the nextplanstartdate is less than previous planenddate. Very simple. I just have to compare both. Compare means Lead and Lag.

Next

```
select E.*, Lead(PLANSTARTDATE,1) over(partition by empid order by PLANSTARTDATE) next_start_date from emp E
```

EMPID	PLANCD	PLANSTARTDATE	PLANENDDATE	NEXT_START_DATE
101	A	01-JAN-24	31-JUL-24	01-APR-24
101	B	01-APR-24	31-DEC-24	-
102	A	01-JAN-24	31-DEC-24	01-JAN-25
102	B	01-JAN-25	31-JUL-25	-
103	A	01-JAN-24	31-JUL-24	01-AUG-24
103	B	01-AUG-24	31-MAR-25	01-JAN-25
103	C	01-JAN-25	31-JUL-25	-

Boundary

Now just compare the condition and get the data.

```
with N1 as(  
select E.*, Lead(PLANSTARTDATE,1) over(partition by empid order by PLANSTARTDATE) next_start_date from emp E  
) select * from n1 where NEXT_START_DATE < PLANENDDATE
```

EMPID	PLANCD	PLANSTARTDATE	PLANENDDATE	NEXT_START_DATE
101	A	01-JAN-24	31-JUL-24	01-APR-24
103	B	01-AUG-24	31-MAR-25	01-JAN-25

2 rows selected.

condition

got it

LIKE OPERATOR

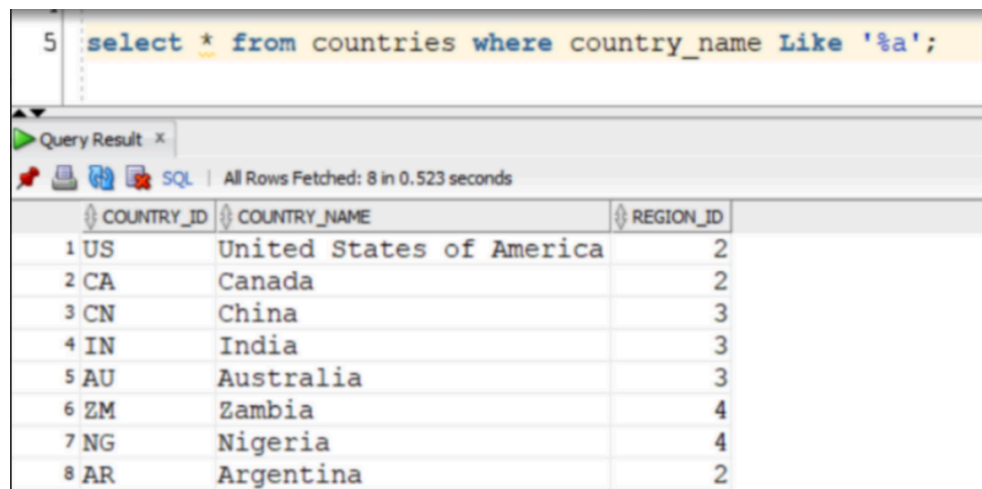
Like operator is used for pattern searching. used as wild card searches.

select * from countries where country_name Like 'A%'; -- it means country_name must start with "A", does not matter what coming after A, but it must start with "A"

select * from countries where country_name Like '%a'; -- it means country_name must end with "a", does not matter what before a, but it must end with "a"

select * from countries where country_name Like '%a%'; -- it means must contain "a", it may comes first, or last or in between, any where

--> Like operator is case sensitive.(Only on Oracle, not in MSSQL)



COUNTRY_ID	COUNTRY_NAME	REGION_ID
1	US	2
2	CA	2
3	CN	3
4	IN	3
5	AU	3
6	ZM	4
7	NG	4
8	AR	2

All ending with 'a'

(only in ORACLE)

Oracle case sensitive
↗

```
select * from hr.countries where COUNTRY_NAME like '%A'
```

no data found

↓
Because NO ONE
ending with
capital A.

--> The same query will not give us any output, because none of the country name ending with capital "A" so, as LIKE operator is case sensitive, we have to match the exact case. small letter to small letter and capital letter to capital letter.

--> `select * from countries where lower(country_name) Like '%a';`

Here In above sql, I am converting this country name in lower case, and now it will match with like '%a'

```
select * from hr.countries where upper(COUNTRY_NAME) like '%A'
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AR	Argentina	2
AU	Australia	3
CA	Canada	2
CN	China	3
IN	India	3
ML	Malaysia	3
NG	Nigeria	4
US	United States of America	2

ARGENTINA

```
select * from hr.countries where COUNTRY_NAME like '%a%a%'
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AU	Australia	3
CA	Canada	2
JP	Japan	3
ML	Malaysia	3
US	United States of America	2
ZM	Zambia	4

2 times
a

All Emp who joined in June.

```
select * from hr.employees where HIRE_DATE like '%JUN%'
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE
100	Steven	King	SKING	515.123.4567	17-JUN-03
105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05
133	Jason	Mallin	JMALLIN	650.127.1934	14-JUN-04
182	Martha	Sullivan	MSULLIVA	650.507.9878	21-JUN-07

Like come with OR and AND operators and behave the same.

```
select * from hr.countries where country_name like '%c%' or country_name like '%a%'
```

This above query will give you all the country_name with contains "c" OR contains "a", either one character or both character

```
1 select * from hr.countries where country_name like '%c%' or country_name like '%a%'
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AR	Argentina	2
AU	Australia	3
BR	Brazil	2
CA	Canada	2
CH	Switzerland	1
CN	China	3
DE	Germany	1
DK	Denmark	1
FR	France	1

either c
or
a

```
select * from hr.countries where country_name like '%c%' And country_name like '%a%'
```

```
select * from hr.countries where country_name like '%c%' and country_name like '%a%'
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
FR	France	1
US	United States of America	2

2 rows selected.

Both must be
there
C & a

NOT LIKE

Just like we have LIKE operator , which is used for pattern matching we have "not like"

--> Give me all country_name which does not contains "C"

```
1 select * from hr.countries where country_name not like '%c%'
```

not like

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AR	Argentina	2
AU	Australia	3
BE	Belgium	1
BR	Brazil	2
CA	Canada	2

How to check if second character is "t" in countries table:

select * from countries where country_name Like '_t%';

```
select * from hr.countries where country_name like '_t%'
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
IT	Italy	1

↓
1 time
underscore
means
1 place is
fix

If I want 3rd character as "t"

select * from countries where country_name Like '__t%';

In this query , we put ' _t%'. first _ (underscore) means, does not ,matter what we have on 1st position, we reserve that and we only take care of 2nd position and after that we have % means, does not matter what comes after 't'.

```
select * from hr.countries where country_name like ' _t%'
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
NL	Netherlands	1

→ 2 underscore

```
select * from hr.countries where country_name like ' __t%'
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AU	Australia	3
CH	Switzerland	1
UK	United Kingdom	1
US	United States of America	2

→ 3 underscore

4 rows selected.